

Setup

This report will compare the success of the agent in a simple geometry with two evenly spaced targets over a variety of parameter values for four parameters: h_o (the attraction of the targets), h_b the base attraction of the targets, σ (how wide the agent's attraction to the target is), and β (inverse noise). For each parameter, we will vary two other aspects of the simulation and hold all other parameters constant: the way the agent encodes position and the symmetry of the target's attraction.

In the even attraction case, we will define success as the minimum distance to a target over the course of the simulation divided by the sum of the distances at that time. In the uneven attraction case we will define success as the minimum distance to the more attractive target over the course of the simulation divided by the sum of the distances at that time. Because the agent starts out evenly spaced between the target, the maximum value for success (the worst possible value) is 0.5. For each set of simulation settings we will take 50 samples and report the average success score for those settings.

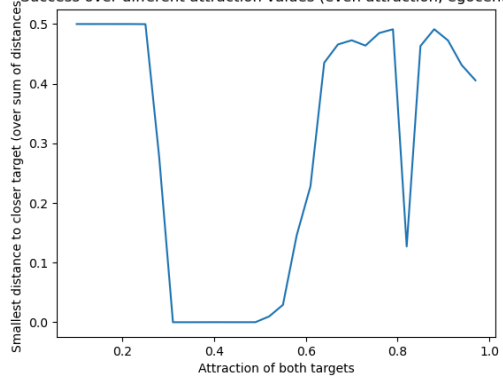
Parameter values:

Parameter	Constant	Change	Number of values
h_o	[0.45,0.45] (even case) [0.45,0.46] (uneven case)	[0.1,0.1]->[0.97,0.97] [0.1,0.11]->[0.97,0.98]	30, increments of 0.03
h_b	0.2	0->0.5	20, increments of ~0.026
σ	0.5	0.1->0.6	20, increments of ~0.026
β	100	20->120	30, increments of ~3.45

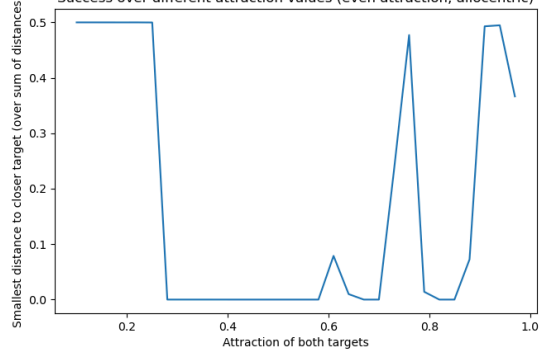
h_o

In all of the situations we looked at, the success value was very close to 0.5 for the smallest h_o values we looked at, up until a little past 0.25, which makes sense considering that we would expect targets to have to exhibit some baseline level of attractiveness to get the agent to move to them. We see that in both egocentric cases that we start to see more failure around 0.6. This appears to be because at those values the agent starts to drift away from the target in the x direction before the agent reaches a target. In the even attraction, allocentric case, the spike in failure around 0.75 seems attributable to a very late decision.

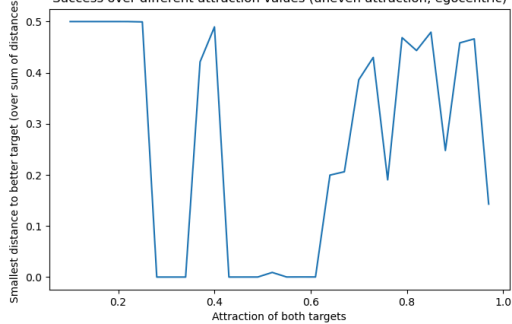
Success over different attraction values (even attraction, egocentric)



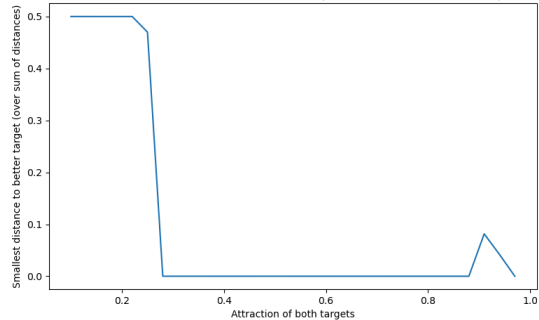
Success over different attraction values (even attraction, allocentric)



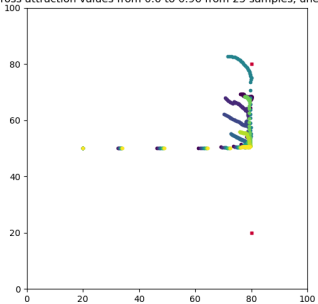
Success over different attraction values (uneven attraction, egocentric)



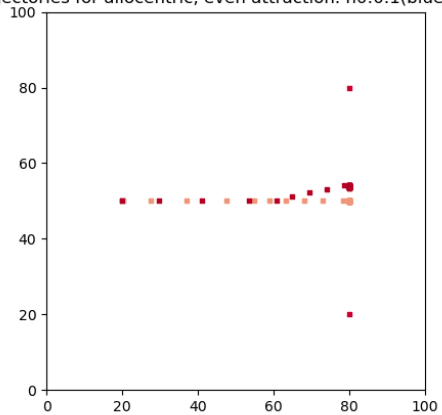
Success over different attraction values (uneven attraction, allocentric)



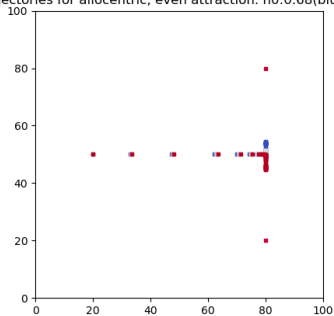
Average trajectories across attraction values from 0.6 to 0.96 from 25 samples, uneven attraction, egocentric



Mean trajectories for allocentric, even attraction. $h_0: 0.1(\text{blue}) \rightarrow 0.35(\text{red})$

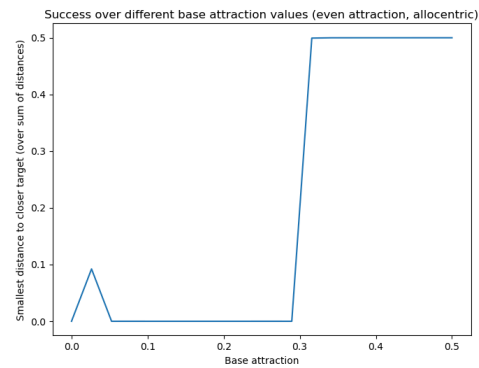
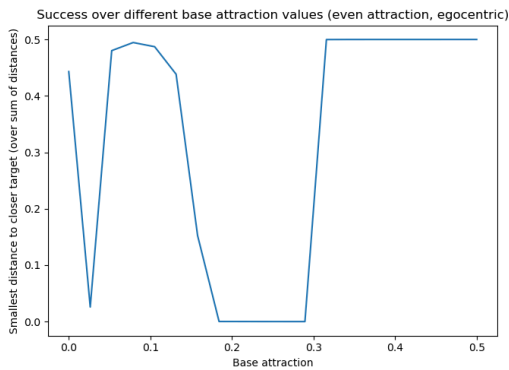


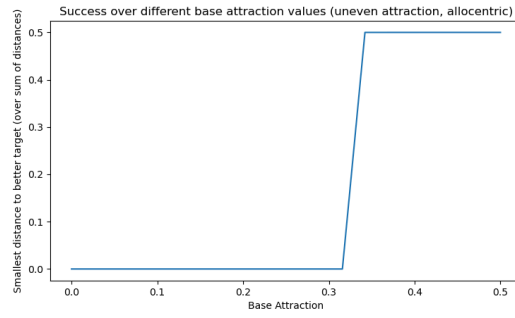
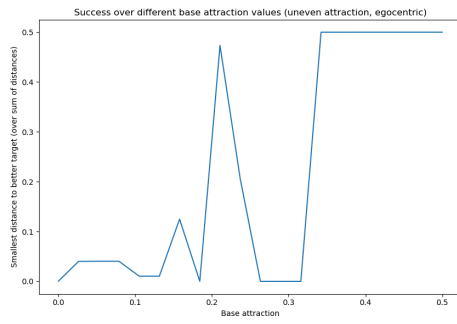
Mean trajectories for allocentric, even attraction. $h_0: 0.68(\text{blue}) \rightarrow 0.8(\text{red})$



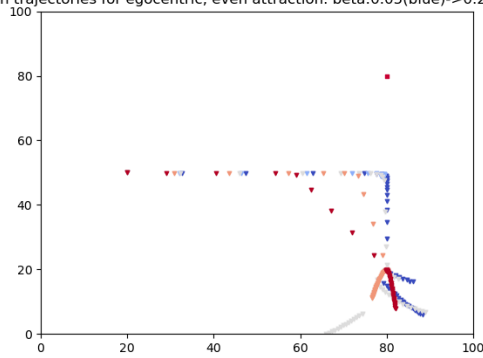
h_b

In all cases, we see that for the highest h_b values in our range (all cases starting around 0.3), we see success values close to 0.5, which, along with some analysis of the trajectories at these values, indicates that the agent fails to reach any target. In the allocentric case, we see that the agent is generally successful across all smaller h_b values. In the egocentric case, we see spikes at different points. In the case of even attraction, we see that from h_b values of around 0.05 to 0.2 there is a large spike in failure. Based on some of the observed trajectories, it appears that in these cases the agent is often failing to make a decision, or is making a very late decision and doesn't have time to reach the target despite heading in that direction. We also see the agent wander a lot more in those trajectories once reaching a target. In the uneven attraction case, we see that there is a smaller, acute spike in failure around 0.22. Based on the analysis of trajectories it seems that this is due to reaching the less attractive target all the time.

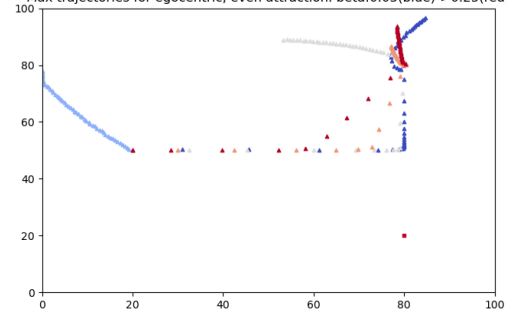




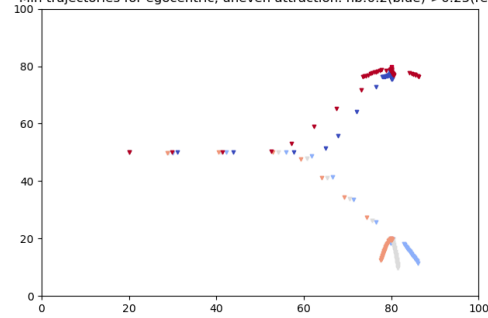
Min trajectories for egocentric, even attraction. beta:0.05(blue)->0.25(red)



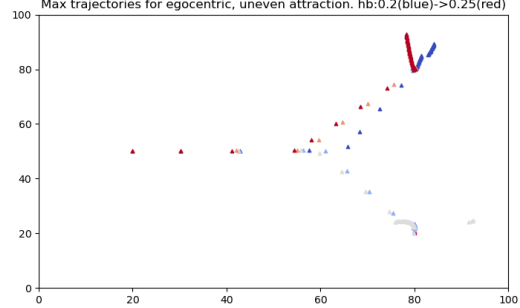
Max trajectories for egocentric, even attraction. beta:0.05(blue)->0.25(red)



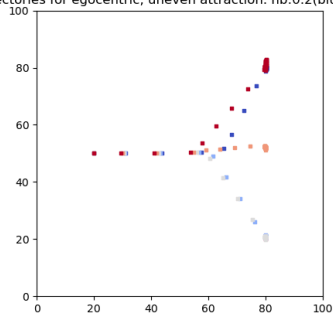
Min trajectories for egocentric, uneven attraction. hb:0.2(blue)->0.25(red)



Max trajectories for egocentric, uneven attraction. hb:0.2(blue)->0.25(red)

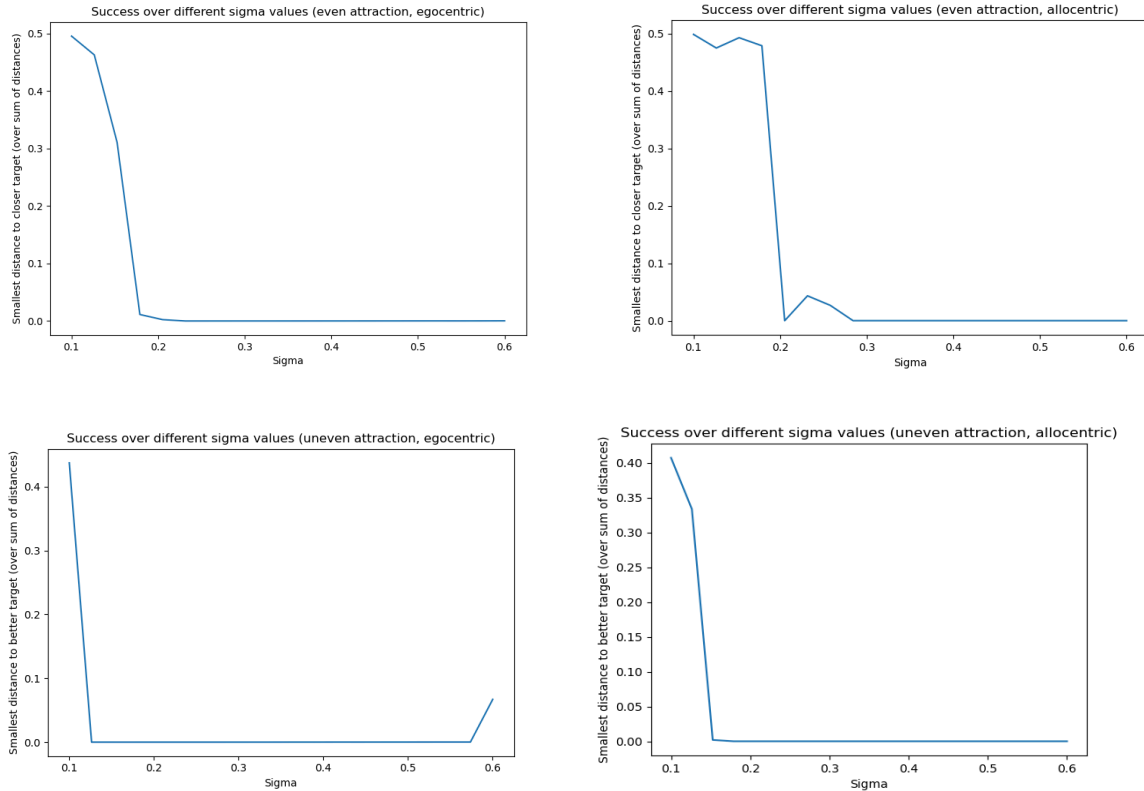


Mean trajectories for egocentric, uneven attraction. hb:0.2(blue)->0.25(red)



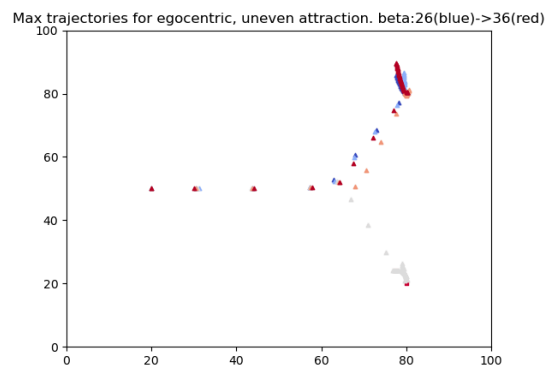
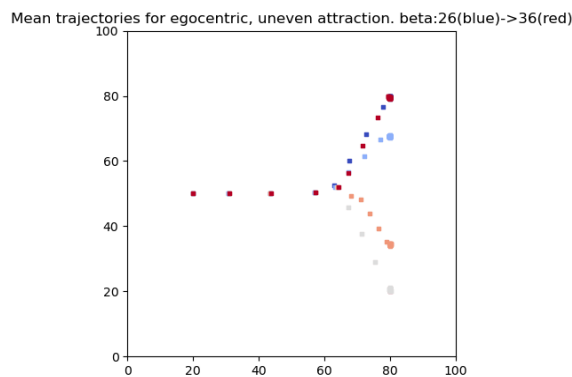
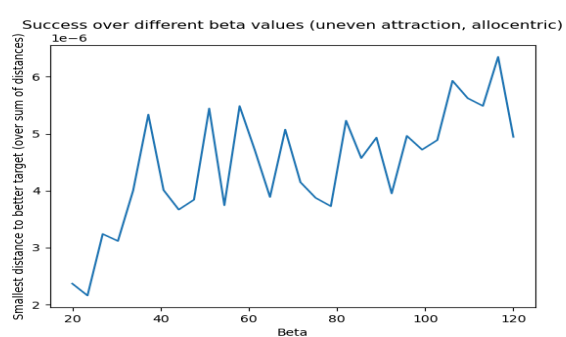
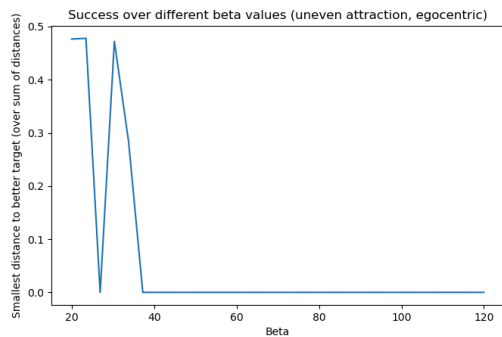
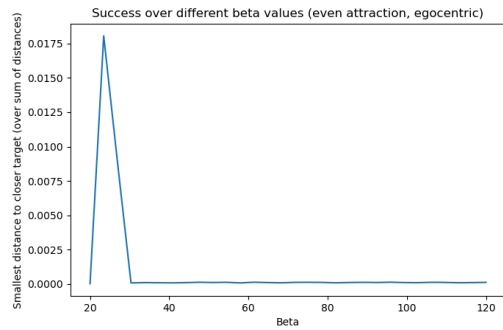
Sigma

Generally, we see that at lower values of sigma the agent has a success score around 0.5, following which there is a steep drop off (anywhere from 0.11->0.2) where the success goes to very close to 0 and stays there. In the egocentric and uneven attraction cases the success increases at a lower sigma value, which is consistent with what we would expect.



Beta

In the allocentric case we see very low success values in both the even and uneven attraction cases (note the y-axis scale), indicating that under these conditions the agent is both successful at reaching a target when there is no better option, and is successful at choosing the better option when there is one. In the egocentric, even attraction case we also see a maximum around 0.0175, which indicates that in the vast majority of cases the agent successfully reaches a target. In the uneven case, we see a spike around beta values of 30, which, after looking at the trajectories, appears to be the result of the agent choosing the wrong target. We can see that when beta is 31 the agent never reaches the correct target in the max trajectory plot.



Base parameter values for other parameters

N=100

L=100

T=5000

periodicflag=0

rEgo=0

rEgoTarget=0

Egonumber=1

distf=1

$adistf=1$

$dt=0.1$

$v0=0.05$

$vot=[0,0]$

Theta and J have same setup ($\nu=0.5$)

Geometry:

Agent initial: (20,50), target1 initial: (20,80), target2 initial: (80,80)